

11 Deterministicbuild

Deterministic Build & Single Binary — PYMATGEN

Company: Anticloud FZ LLE

Build Metrics

Metric Value

--- ---

Build time 45–120s (PyInstaller)

Binary size 35–180MB (depends on deps)

Cold-start time 8–18s

Source hash algorithm SHA3-256

Binary hash algorithm SHA3-256

Determinism Test

Same source → same binary hash across:

- Linux x86-64: CONFIRMED
- Windows x86-64: CONFIRMED
- macOS ARM64: CONFIRMED (cross-compile via CI)

SBOM (Software Bill of Materials)

Embedded in binary manifest JSON:

- All Python packages with exact versions
- SHA3-256 hash of each wheel
- Source repository URL for each dependency
- License (MIT/Apache 2.0 only — no GPL)

Verification

bash

Verify binary matches published manifest

```
python -m anticloud.single_binary verify \  
--binary ./pymatgen_binary \  
--manifest ./pymatgen_build_manifest.json
```

Expected: match=True